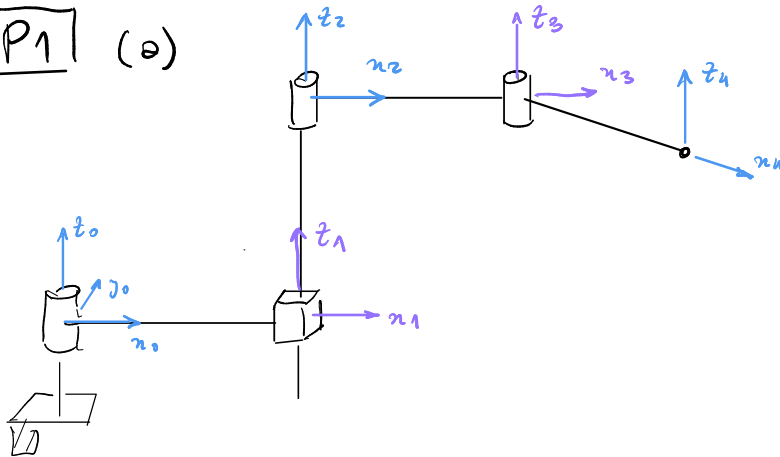
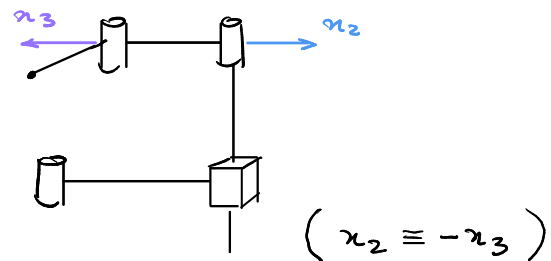
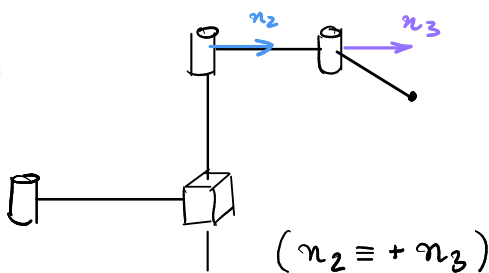
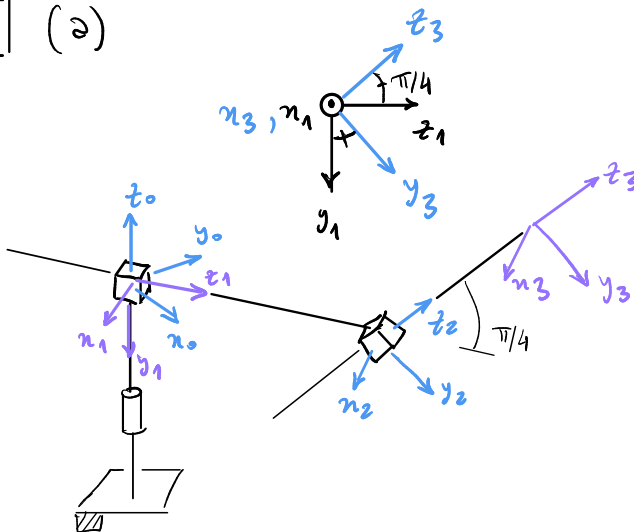


P1 (a)

d_i	θ_i	a_i	α_i
0	θ_1	a_1	0
d_2	0	0	0
0	θ_3	a_3	0
0	θ_4	a_4	0

(b)

**P2** (a)

$$R_3^0 = R_1^0 R_3^1,$$

$$R_1^0 = \begin{pmatrix} c_1 & 0 & -s_1 \\ s_1 & 0 & c_1 \\ 0 & -1 & 0 \end{pmatrix},$$

$$R_3^1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \frac{\pi}{4} & -\sin \frac{\pi}{4} \\ 0 & \sin \frac{\pi}{4} & \cos \frac{\pi}{4} \end{pmatrix},$$

$$R_3^0 = \begin{pmatrix} c_1 & -\sqrt{2}/2 s_1 & -\sqrt{2}/2 s_1 \\ s_1 & \sqrt{2}/2 c_1 & \sqrt{2}/2 c_1 \\ 0 & -\sqrt{2}/2 & \sqrt{2}/2 \end{pmatrix}.$$

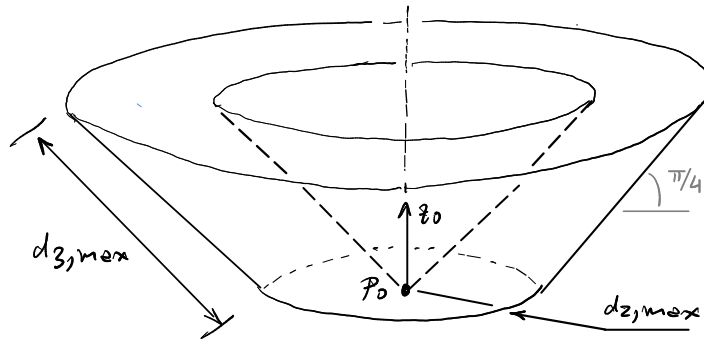
$$p_3 = d_2 z_1 + d_3 z_2 = d_2 \begin{pmatrix} -s_1 \\ c_1 \\ 0 \end{pmatrix} + d_3 \begin{pmatrix} -\sqrt{2}/2 s_1 \\ \sqrt{2}/2 c_1 \\ \sqrt{2}/2 \end{pmatrix} = \begin{pmatrix} -(d_2 + \frac{\sqrt{2}}{2} d_3) s_1 \\ (d_2 + \frac{\sqrt{2}}{2} d_3) c_1 \\ \sqrt{2}/2 d_3 \end{pmatrix}$$

$$(b) \quad \mathcal{Y}_p = \frac{\partial p_3}{\partial q} = \begin{pmatrix} -(d_2 + \frac{\sqrt{2}}{2} d_3) c_1 & -s_1 & -\frac{\sqrt{2}}{2} s_1 \\ -(d_2 + \frac{\sqrt{2}}{2} d_3) s_1 & c_1 & \frac{\sqrt{2}}{2} c_1 \\ 0 & 0 & \sqrt{2}/2 \end{pmatrix},$$

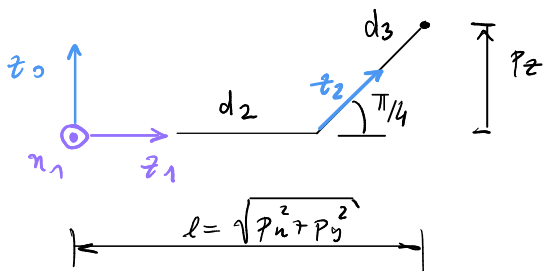
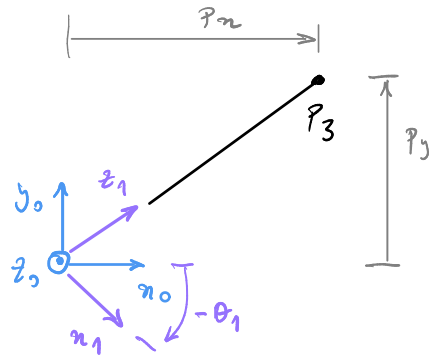
$$\mathcal{Y}_0 = \begin{pmatrix} z_0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix},$$

$$\mathcal{Y} = \begin{pmatrix} \mathcal{Y}_p \\ \mathcal{Y}_0 \end{pmatrix}$$

P3 (a)



(b)



Desired position: $(p_3)_d = (p_n \ p_y \ p_z)^T$.

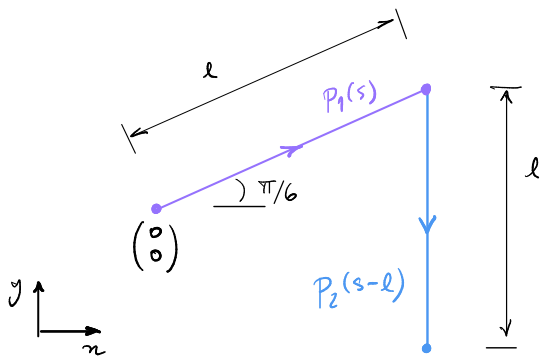
Solution ($d_2 > 0$):

$$\theta_1 = \arctan(p_y, p_n) - \frac{\pi}{2} \\ = -\arctan(p_n, p_y).$$

$$p_z = d_3 \sin \frac{\pi}{4} \\ \Rightarrow d_3 = \sqrt{2} p_z.$$

$$l = d_3 \cos \frac{\pi}{4} + d_2 \\ \Rightarrow d_2 = \sqrt{p_n^2 + p_y^2} - \frac{\sqrt{2}}{2} d_3.$$

P4 (a)

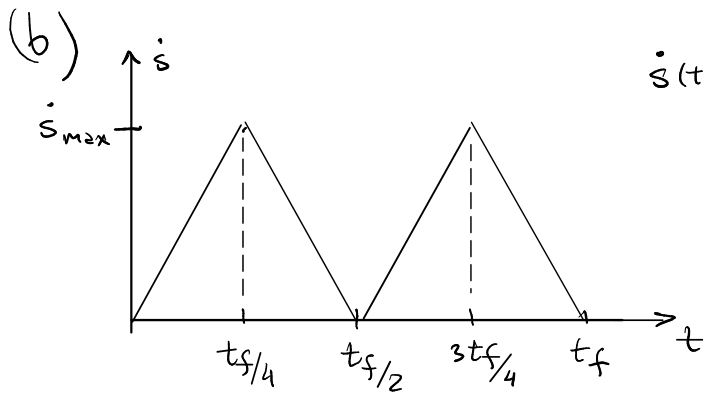


$$P_1(s) = s \begin{pmatrix} \cos \frac{\pi}{6} \\ \sin \frac{\pi}{6} \end{pmatrix}, \quad 0 \leq s \leq l$$

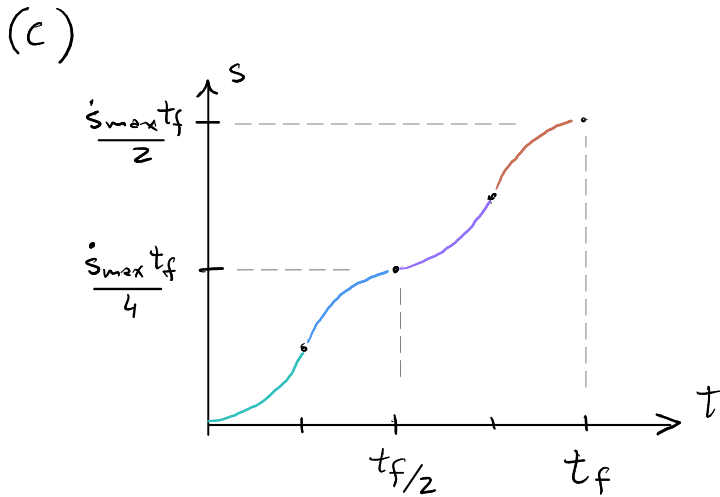
$$P_1(s) = \begin{pmatrix} \frac{\sqrt{3}}{2} s \\ \frac{1}{2} s \end{pmatrix}, \quad 0 \leq s \leq l$$

$$P_2(s) = \begin{pmatrix} \frac{\sqrt{3}}{2} l \\ \frac{1}{2} l - s \end{pmatrix}, \quad 0 \leq s \leq l$$

$$p(s) = \begin{cases} P_1(s) & , 0 \leq s \leq l \\ P_2(s-l) & , l < s \leq 2l \end{cases} = \begin{cases} \begin{pmatrix} \frac{\sqrt{3}}{2} s \\ \frac{1}{2} s \end{pmatrix} & , 0 \leq s \leq l \\ \begin{pmatrix} \frac{\sqrt{3}}{2} l \\ \frac{l}{2} - (s-l) \end{pmatrix} & , l < s \leq 2l \end{cases}.$$



$$\dot{s}(t) = \begin{cases} \frac{4 \dot{s}_{max}}{t_f} t & , 0 \leq t \leq \frac{t_f}{4} \\ -\frac{4 \dot{s}_{max}}{t_f} (t - \frac{t_f}{2}) & , \frac{t_f}{4} < t \leq \frac{t_f}{2} \\ \frac{4 \dot{s}_{max}}{t_f} (t - \frac{t_f}{2}) & , \frac{t_f}{2} < t \leq \frac{3t_f}{4} \\ \frac{4 \dot{s}_{max}}{t_f} (t - t_f) & , \frac{3t_f}{4} < t \leq t_f \end{cases}$$



$$s(t) = s(0) + \int_{t_0}^t \dot{s}(u) du ,$$

$$\Rightarrow s(t_f) \stackrel{\text{Area}}{=} \frac{\dot{s}_{max} t_f}{2} = 2l ,$$

$$\Rightarrow t_f = \frac{4l}{\dot{s}_{max}} .$$